Yonathan Fisseha

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Research Overview

I work at the intersection of programming languages and computer architecture, developing formal foundations for performant, privacy-preserving architectures. My research builds precise instruction-set semantics, mechanizes them in theorem provers, and links them to microarchitectural behavior. I also develop algebraic semantics for emerging HDLs to give their abstractions a more compositional foundation. In parallel, I maintain a large empirical study of SAT-solver heuristics which are the backends for many reasoning frameworks. Beyond the University of Michigan, I collaborated with Intel's GPU architecture and compiler team to formalize Intel's new GPU ISA's memory model and its sub-workgroup IFP execution semantics.

Research

Formal Verification of Low-trust Architectures 2023
Q. Tan, ***Y. Fisseha, *S. Chen, L. Biernacki, J.B. Jeannin, S. Malik, T. Austin* *CCS*

We provide an end-to-end security formal verification for a low-trust architecture. Our proof is on two levels: ISA-level semantics proofs that are followed by implementation-level model checking proofs.

Distinguished paper award winner at CCS'23. * = co-first author.

HashStep: Data and Computation Integrity Enforcement Under Submission
***Y. Fisseha**, Q. Tan, S. Chen, M. Demissie, L. Biernacki, J.B. Jeannin, S. Malik, T. Austin* *Under Submission*

We build on our prior work to provide confidentiality of data, integrity of data, and integrity of computation. We formalize the instruction set's extension as a smallstep semantics and mechanize our main security result in the Lean Theorem Prover.

HardKAT Present
***Y. Fisseha**, S. Chen, J.B. Jeannin, T. Austin* *In Progress*

HardKAT is a formally defined language for accelerator-oriented designs. It is built on the algebraic framework Partially Concurrent Kleene Algebra (POCKA). We show that the semantics is sound and are finalizing a completeness proof. A version of this paper was presented in *TechCon 2022*.

Tabular Sat Present
Y. Fisseha*, J. Chen*, M. Fleury*, A. Biere, S. Karem *In Progress*

We start with a blank slate and carefully categorize various techniques and build very simple solvers that combine these categories. We are running various large-scale performance tests on recent benchmarks from SAT competitions. "Revisiting Clause Vivification" presents an implementation improvement based on our experiments; presented in the POS'25 workshop by our team members. * = co-first author.

Twine: A Chisel Extension for Component-level Heterogeneous Design 2022
*S. Chen **Y. Fisseha**, J.B. Jeannin, T. Austin* *DATE*

Twine is an extension to the Chisel hardware generator language. It improves re-usability and composability of hardware components, allowing designers to stay productive in heterogeneous design. It provides well defined interfaces and automates control-flow communication and data type and bandwidth conversions between components implementing these interfaces. This work has appeared in DATE'22 as Twine and TechCon 2021 as SimpleChisel.

Work Experience

Intel, GPU Compilers Team 2025
Research Intern *Santa Clara, CA*

I formalized a subset of the execution model for the next generation Intel GPU ISA. I used Alloy6, particularly its recently added temporal logic features, to model the execution of programs under a more permissive sub-workgroup progress model. The implementation serves as a first step in the formalization process and also a formal ground-truth for other tools, such as internal simulators.

Intel, GPU Compilers Team 2024
Research Intern *Santa Clara, CA*

I formalized the memory model for Intel's next generation GPU ISA. The language was modified as various limitations were discovered in the process of formalizing it and the work culminated in a soundness proof for a version of the memory consistency model.

Microsoft Intune 2018
Software Engineer Intern *San Jose, CA*

I designed and implemented Network-Fencing and Geo-Fencing testing tools for the Intune Android client using Hyper-V and Android Mock Locations. The testing tool was integrated into the existing CI infrastructure allowing developers to automatically test code that was previously tested manually.

Impellia 2016-17
Software Developer *Remote*

I worked with the startup implementing multiple projects that leverage algorithms related to athletics designed and licensed by various universities. The projects helped validate the algorithms for future products.

University of Colorado School of Medicine 2015-17
Software Developer *Aurora, CO*

I worked with doctors and researchers to quickly validate clinical product concepts through prototypes and user testing. The products involved tracking patient data, numerical analysis, billing, and hour tracking.

Non-peer Reviewed Reports

Migrating Browser Automation Programs for End-to-End Testing 2021
Y. Fisseha, *Y. Li* *University of Michigan*

We applied program synthesis techniques supported by a MaxSMT solver to automate a majority of test case migration between version of a web application. We demonstrate how to generate a new Python test program that works on the new version of the frontend given the old version of the frontend HTML and the old Python test program.

Compression-aware Algorithms 2020
Y. Fisseha, *N. Brunelle* *University of Virginia*

I studied if the LZ-family of string-alignment algorithms can be redesigned to be compression aware. We propose algorithms that work on various class of input strings and experimentally characterize the input strings that can be processed by such algorithms.

Secure Data Summarization for Stream Data 2019
Y. Fisseha, *A. Hithnawi* *University of California Berkeley*

IoT data is temporal by nature thus data summarization techniques can be applied to reduce the data size before Partially homomorphic encryption schemes are applied to secure the data moving forward. We proposed a system design that allows secure summarization of streaming data.

Education

University of Michigan 2020-Present
Doctor of Philosophy in Computer Science and Engineering *Ann Arbor, MI*

I am co-advised by Professor Todd Austin and Professor Jean-Baptiste Jeannin. I currently hold a GSRA position.

University of Virginia 2016-2020
Bachelor of Science in Computer Science with High Distinction *Charlottesville, VA*

Teaching Service

Graduate Student Instructor 2024, 2026
EECS 573: Microarchitectures, University of Michigan *Ann Arbor*

I was the student instructor for EECS 573. I drafted the exam, advised students on final projects, managed and graded in-class presentations.

TEALS Volunteer Teacher 2022
Fenton High School *Remote*

I was one of the teaching volunteers at Fenton High School. We led a class of 16 students. This is an AP 1 Computer Science full year class using Java.

Lead TEALS Volunteer Teacher 2021
Eppler Junior High School *Remote*

I was the lead volunteer instructor for a class of about 15 students. A full time teacher was in the classroom providing in person support while I gave the lectures and helped students with projects. Languages: Snap and Python.

Teaching Assistant 2018-2020
University of Virginia *Charlottesville, VA*

I was an undergraduate teaching assistant for Algorithms, Advanced Software Development, Programming Languages for Web Applications, and Operating Systems.

Awards & Honors

Jane Street Fellowship Finalist 2024
Jane Street Capital
ACM CCS Distinguished Paper Award 2023
CCS
Rackham Merit Fellowship 2020
University of Michigan
University Honors 2019, 2020
University of Virginia
QuestBridge Scholar 2016
QuestBridge
Daniels Scholar 2016
Daniels Fund

Citizenship

Status: *United States Citizen*